

An Introduction to MongoDB

MongoDB is an open source NoSQL type of document database. It is a cross-platform database that provides high levels of performance, scalability and can be used in enterprise applications that deal with large sets of data.



MongoDB was created in 2007, and is built on a horizontal scale-out architecture that stores data in a document style described in the JSON language. It can be used on platforms such as Windows and Linux, among others.

Reasons to use MongoDB

MongoDB can handle huge sets of data, is easy and quick to access, and can be used in a variety of applications and settings. It is used in the Internet of Things (IoT), banking and finance, gaming, e-commerce, logistics and content management.

- It is fast and powerful, and convenient for moving data because of its document data model.
- It supports a large set of both data and traffic because of its horizontal, scale-out architecture.
- It offers a great user experience for developers.

- It is accessible to anyone from anywhere — via the cloud through MongoDB Atlas, and free through the community edition.
- It supports rapid iterative development.
- It has great community support.

Problems that MongoDB solves

The problems that MongoDB solves are related to data management and software development.

1. It enables the integration of large amounts of data from different sources into a unified single view, something other databases are unable to do.
2. It is able to describe and support different types of complex data structures, even when they are constantly evolving. Geospatial data is also supported.
3. Unlike other databases, MongoDB is highly scalable, enabling it to support applications that need high

and efficient performance.

4. Because of its flexibility in deployment, it is able to support hybrid and cloud applications, both private and public.
5. On account of its document databases that are easier to use and understand, it supports agile development and collaboration in projects.

Features of MongoDB

Most NoSQL databases have certain default features common to all of them, but MongoDB has some additional useful features.

- MongoDB is a high-performance efficient database. This is because it supports embedded data models and its query operations are faster.
- It supports almost all CRUD operations and has a rich query language.
- Its auto-replication feature provides a good back-up mechanism if a server fails.
- It has the sharding feature that is responsible for horizontal scalability.
- It supports multiple storage engines.

Application areas

MongoDB is mainly used in Web applications as the preferred database. It is used by the MEAN stack (a popular Web development stack) as the main database. MEAN is a combination of MongoDB, AngularJS, ExpressJS and NodeJS.

